

Checking for missing values using isnull()

```
In [33]: # Import pandas and numpy in order to check missing values in PandasDataFrame  
import pandas as pd  
import numpy as np  
import warnings  
warnings.filterwarnings("ignore")
```

```
In [34]: # Load the dataset in dataframe object df  
df = pd.read_csv("Downloads/Students_Performance.csv")
```

```
In [35]: df.shape
```

```
Out[35]: (30, 7)
```

```
In [36]: # Display the data frame  
df
```

Out [36]:

	Gender	Math_Score	Reading_Score	Writing_Score	Placement_Score	Club_Join_Date	Placement_Offer_Coi
0	Male	69.0	95.0	71.0	93	2020	
1	Male	77.0	82.0	78.0	86	2019	
2	Male	76.0	90.0	70.0	83	2018	
3	Male	63.0	88.0	78.0	77	2021	
4	Female	70.0	76.0	63.0	98	2019	
5	Male	68.0	78.0	79.0	79	2021	
6	Male	79.0	95.0	NaN	89	2018	
7	Male	77.0	NaN	70.0	84	2018	
8	Female	NaN	85.0	68.0	81	2021	
9	Female	62.0	76.0	74.0	98	2021	
10	Female	76.0	88.0	61.0	89	2021	
11	Female	72.0	81.0	78.0	82	2021	
12	Male	78.0	81.0	64.0	80	2018	
13	Female	62.0	76.0	NaN	94	2018	
14	Female	NaN	76.0	66.0	86	2021	
15	Female	79.0	91.0	67.0	79	2018	
16	Male	75.0	85.0	79.0	79	2020	
17	Female	74.0	91.0	73.0	88	2020	
18	Female	60.0	NaN	64.0	77	2021	
19	Male	68.0	78.0	77.0	90	2021	
20	Female	69.0	82.0	63.0	84	2018	
21	Male	NaN	86.0	NaN	85	2021	
22	Male	75.0	95.0	75.0	84	2018	
23	Male	74.0	86.0	66.0	93	2021	
24	Female	80.0	NaN	62.0	95	2020	
25	Female	68.0	83.0	79.0	79	2018	
26	Female	79.0	75.0	71.0	76	2021	
27	Female	73.0	79.0	70.0	100	2021	
28	Male	63.0	77.0	67.0	93	2020	
29	Female	78.0	84.0	74.0	91	2020	

In [37]:

```
df.describe()
```

Out[37]:	Math_Score	Reading_Score	Writing_Score	Placement_Score	Club_Join_Date	Placement_Offer_Count
count	27.000000	27.000000	27.000000	30.000000	30.000000	30.000000
mean	72.000000	83.666667	70.629630	86.400000	2019.766667	2.000000
std	6.101702	6.342773	5.917043	6.926212	1.304722	0.787839
min	60.000000	75.000000	61.000000	76.000000	2018.000000	1.000000
25%	68.000000	78.000000	66.000000	80.250000	2018.000000	1.000000
50%	74.000000	83.000000	70.000000	85.500000	2020.000000	2.000000
75%	77.000000	88.000000	76.000000	92.500000	2021.000000	3.000000
max	80.000000	95.000000	79.000000	100.000000	2021.000000	3.000000

In [38]: `df.info()`

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 30 entries, 0 to 29
Data columns (total 7 columns):
#   Column                Non-Null Count  Dtype
---  -
0   Gender                30 non-null    object
1   Math_Score            27 non-null    float64
2   Reading_Score         27 non-null    float64
3   Writing_Score         27 non-null    float64
4   Placement_Score       30 non-null    int64
5   Club_Join_Date        30 non-null    int64
6   Placement_Offer_Count 30 non-null    int64
dtypes: float64(3), int64(3), object(1)
memory usage: 1.8+ KB
```

In [39]: *# Use isnull() function to check null values in the dataset.*
`df.isnull()`

Out[39]:

	Gender	Math_Score	Reading_Score	Writing_Score	Placement_Score	Club_Join_Date	Placement_Offer_Cou
0	False	False	False	False	False	False	Fa
1	False	False	False	False	False	False	Fa
2	False	False	False	False	False	False	Fa
3	False	False	False	False	False	False	Fa
4	False	False	False	False	False	False	Fa
5	False	False	False	False	False	False	Fa
6	False	False	False	True	False	False	Fa
7	False	False	True	False	False	False	Fa
8	False	True	False	False	False	False	Fa
9	False	False	False	False	False	False	Fa
10	False	False	False	False	False	False	Fa
11	False	False	False	False	False	False	Fa
12	False	False	False	False	False	False	Fa
13	False	False	False	True	False	False	Fa
14	False	True	False	False	False	False	Fa
15	False	False	False	False	False	False	Fa
16	False	False	False	False	False	False	Fa
17	False	False	False	False	False	False	Fa
18	False	False	True	False	False	False	Fa
19	False	False	False	False	False	False	Fa
20	False	False	False	False	False	False	Fa
21	False	True	False	True	False	False	Fa
22	False	False	False	False	False	False	Fa
23	False	False	False	False	False	False	Fa
24	False	False	True	False	False	False	Fa
25	False	False	False	False	False	False	Fa
26	False	False	False	False	False	False	Fa
27	False	False	False	False	False	False	Fa
28	False	False	False	False	False	False	Fa
29	False	False	False	False	False	False	Fa

In [40]:

df.isnull().sum()

Out[40]:

Gender0

Math_Score3

Reading_Score3

Writing_Score3

Placement_Score0

Club_Join_Date0

Placement_Offer_Count0

dtype: int64

In [41]:

```
# To create a series true for NaN values for specific columns.  
series = pd.isnull(df["Math_Score"])  
df[series]
```

Out[41]:

	Gender	Math_Score	Reading_Score	Writing_Score	Placement_Score	Club_Join_Date	Placement_Offer_Coi
8	Female	NaN	85.0	68.0	81	2021	
14	Female	NaN	76.0	66.0	86	2021	
21	Male	NaN	86.0	NaN	85	2021	

Checking for missing values using notnull()

In [42]:

```
# Use notnull() function to check null values in the dataset.  
df.notnull()
```

Out[42]:

	Gender	Math_Score	Reading_Score	Writing_Score	Placement_Score	Club_Join_Date	Placement_Offer_Cou
0	True	True	True	True	True	True	True
1	True	True	True	True	True	True	True
2	True	True	True	True	True	True	True
3	True	True	True	True	True	True	True
4	True	True	True	True	True	True	True
5	True	True	True	True	True	True	True
6	True	True	True	False	True	True	True
7	True	True	False	True	True	True	True
8	True	False	True	True	True	True	True
9	True	True	True	True	True	True	True
10	True	True	True	True	True	True	True
11	True	True	True	True	True	True	True
12	True	True	True	True	True	True	True
13	True	True	True	False	True	True	True
14	True	False	True	True	True	True	True
15	True	True	True	True	True	True	True
16	True	True	True	True	True	True	True
17	True	True	True	True	True	True	True
18	True	True	False	True	True	True	True
19	True	True	True	True	True	True	True
20	True	True	True	True	True	True	True
21	True	False	True	False	True	True	True
22	True	True	True	True	True	True	True
23	True	True	True	True	True	True	True
24	True	True	False	True	True	True	True
25	True	True	True	True	True	True	True
26	True	True	True	True	True	True	True
27	True	True	True	True	True	True	True
28	True	True	True	True	True	True	True
29	True	True	True	True	True	True	True

In [43]:

df.notnull().sum()

Out[43]:

Gender30

Math_Score27

Reading_Score27

Writing_Score27

Placement_Score30

Club_Join_Date30

Placement_Offer_Count30

dtype: int64

In [44]:

```
# To create a series true for NaN values for specific columns.  
series1 = pd.notnull(df["Math_Score"])  
df[series1]
```

Out[44]:

	Gender	Math_Score	Reading_Score	Writing_Score	Placement_Score	Club_Join_Date	Placement_Offer_Co
0	Male	69.0	95.0	71.0	93	2020	
1	Male	77.0	82.0	78.0	86	2019	
2	Male	76.0	90.0	70.0	83	2018	
3	Male	63.0	88.0	78.0	77	2021	
4	Female	70.0	76.0	63.0	98	2019	
5	Male	68.0	78.0	79.0	79	2021	
6	Male	79.0	95.0	NaN	89	2018	
7	Male	77.0	NaN	70.0	84	2018	
9	Female	62.0	76.0	74.0	98	2021	
10	Female	76.0	88.0	61.0	89	2021	
11	Female	72.0	81.0	78.0	82	2021	
12	Male	78.0	81.0	64.0	80	2018	
13	Female	62.0	76.0	NaN	94	2018	
15	Female	79.0	91.0	67.0	79	2018	
16	Male	75.0	85.0	79.0	79	2020	
17	Female	74.0	91.0	73.0	88	2020	
18	Female	60.0	NaN	64.0	77	2021	
19	Male	68.0	78.0	77.0	90	2021	
20	Female	69.0	82.0	63.0	84	2018	
22	Male	75.0	95.0	75.0	84	2018	
23	Male	74.0	86.0	66.0	93	2021	
24	Female	80.0	NaN	62.0	95	2020	
25	Female	68.0	83.0	79.0	79	2018	
26	Female	79.0	75.0	71.0	76	2021	
27	Female	73.0	79.0	70.0	100	2021	
28	Male	63.0	77.0	67.0	93	2020	
29	Female	78.0	84.0	74.0	91	2020	

In [45]:

```
# See that there are also categorical values in the dataset, for this, you  
# need to use Label Encoding  
from sklearn.preprocessing import LabelEncoder  
  
# Create a LabelEncoder instance  
le = LabelEncoder()
```

```
# Apply the LabelEncoder to the 'gender' column and assign the result back to  
the 'gender' column  
df['Gender'] = le.fit_transform(df['Gender'])  
  
# Optionally, create a new DataFrame with the updated 'gender' column  
newdf = df  
  
# Display the DataFrame with the encoded gender column  
newdf
```


Out[45]:		Gender	Math_Score	Reading_Score	Writing_Score	Placement_Score	Club_Join_Date	Placement_Offer_Coi
	0	1	69.0	95.0	71.0	93	2020	
	1	1	77.0	82.0	78.0	86	2019	
	2	1	76.0	90.0	70.0	83	2018	
	3	1	63.0	88.0	78.0	77	2021	
	4	0	70.0	76.0	63.0	98	2019	
	5	1	68.0	78.0	79.0	79	2021	
	6	1	79.0	95.0	NaN	89	2018	
	7	1	77.0	NaN	70.0	84	2018	
	8	0	NaN	85.0	68.0	81	2021	
	9	0	62.0	76.0	74.0	98	2021	
	10	0	76.0	88.0	61.0	89	2021	
	11	0	72.0	81.0	78.0	82	2021	
	12	1	78.0	81.0	64.0	80	2018	
	13	0	62.0	76.0	NaN	94	2018	
	14	0	NaN	76.0	66.0	86	2021	
	15	0	79.0	91.0	67.0	79	2018	
	16	1	75.0	85.0	79.0	79	2020	
	17	0	74.0	91.0	73.0	88	2020	
	18	0	60.0	NaN	64.0	77	2021	
	19	1	68.0	78.0	77.0	90	2021	
	20	0	69.0	82.0	63.0	84	2018	
	21	1	NaN	86.0	NaN	85	2021	
	22	1	75.0	95.0	75.0	84	2018	
	23	1	74.0	86.0	66.0	93	2021	
	24	0	80.0	NaN	62.0	95	2020	
	25	0	68.0	83.0	79.0	79	2018	
	26	0	79.0	75.0	71.0	76	2021	
	27	0	73.0	79.0	70.0	100	2021	
	28	1	63.0	77.0	67.0	93	2020	
	29	0	78.0	84.0	74.0	91	2020	

Filling missing values using dropna(), fillna(), replace()

```
In [46]: # For replacing null values with NaN
missing_values = ["Na", "na"]
df = pd.read_csv("Downloads/Students_Performance.csv", na_values =
missing_values)
df
```

Out[46]:

	Gender	Math_Score	Reading_Score	Writing_Score	Placement_Score	Club_Join_Date	Placement_Offer_Coi
0	Male	69.0	95.0	71.0	93	2020	
1	Male	77.0	82.0	78.0	86	2019	
2	Male	76.0	90.0	70.0	83	2018	
3	Male	63.0	88.0	78.0	77	2021	
4	Female	70.0	76.0	63.0	98	2019	
5	Male	68.0	78.0	79.0	79	2021	
6	Male	79.0	95.0	NaN	89	2018	
7	Male	77.0	NaN	70.0	84	2018	
8	Female	NaN	85.0	68.0	81	2021	
9	Female	62.0	76.0	74.0	98	2021	
10	Female	76.0	88.0	61.0	89	2021	
11	Female	72.0	81.0	78.0	82	2021	
12	Male	78.0	81.0	64.0	80	2018	
13	Female	62.0	76.0	NaN	94	2018	
14	Female	NaN	76.0	66.0	86	2021	
15	Female	79.0	91.0	67.0	79	2018	
16	Male	75.0	85.0	79.0	79	2020	
17	Female	74.0	91.0	73.0	88	2020	
18	Female	60.0	NaN	64.0	77	2021	
19	Male	68.0	78.0	77.0	90	2021	
20	Female	69.0	82.0	63.0	84	2018	
21	Male	NaN	86.0	NaN	85	2021	
22	Male	75.0	95.0	75.0	84	2018	
23	Male	74.0	86.0	66.0	93	2021	
24	Female	80.0	NaN	62.0	95	2020	
25	Female	68.0	83.0	79.0	79	2018	
26	Female	79.0	75.0	71.0	76	2021	
27	Female	73.0	79.0	70.0	100	2021	
28	Male	63.0	77.0	67.0	93	2020	
29	Female	78.0	84.0	74.0	91	2020	

Filling null values with a single value

In [47]:

```
# filling missing value using fillna()
df = pd.read_csv("Downloads/Students_Performance.csv")
ndf = df
ndf.fillna(0)
```

Out[47]:

	Gender	Math_Score	Reading_Score	Writing_Score	Placement_Score	Club_Join_Date	Placement_Offer_Coi
0	Male	69.0	95.0	71.0	93	2020	
1	Male	77.0	82.0	78.0	86	2019	
2	Male	76.0	90.0	70.0	83	2018	
3	Male	63.0	88.0	78.0	77	2021	
4	Female	70.0	76.0	63.0	98	2019	
5	Male	68.0	78.0	79.0	79	2021	
6	Male	79.0	95.0	0.0	89	2018	
7	Male	77.0	0.0	70.0	84	2018	
8	Female	0.0	85.0	68.0	81	2021	
9	Female	62.0	76.0	74.0	98	2021	
10	Female	76.0	88.0	61.0	89	2021	
11	Female	72.0	81.0	78.0	82	2021	
12	Male	78.0	81.0	64.0	80	2018	
13	Female	62.0	76.0	0.0	94	2018	
14	Female	0.0	76.0	66.0	86	2021	
15	Female	79.0	91.0	67.0	79	2018	
16	Male	75.0	85.0	79.0	79	2020	
17	Female	74.0	91.0	73.0	88	2020	
18	Female	60.0	0.0	64.0	77	2021	
19	Male	68.0	78.0	77.0	90	2021	
20	Female	69.0	82.0	63.0	84	2018	
21	Male	0.0	86.0	0.0	85	2021	
22	Male	75.0	95.0	75.0	84	2018	
23	Male	74.0	86.0	66.0	93	2021	
24	Female	80.0	0.0	62.0	95	2020	
25	Female	68.0	83.0	79.0	79	2018	
26	Female	79.0	75.0	71.0	76	2021	
27	Female	73.0	79.0	70.0	100	2021	
28	Male	63.0	77.0	67.0	93	2020	
29	Female	78.0	84.0	74.0	91	2020	

In [48]:

```
# filling missing values using mean, median and standard deviation of that column.
df = pd.read_csv("Downloads/Students_Performance.csv")
df['Math_Score'] = df['Math_Score'].fillna(df['Math_Score'].mean())
df
```

Out[48]:

	Gender	Math_Score	Reading_Score	Writing_Score	Placement_Score	Club_Join_Date	Placement_Offer_Coi
0	Male	69.0	95.0	71.0	93	2020	
1	Male	77.0	82.0	78.0	86	2019	
2	Male	76.0	90.0	70.0	83	2018	
3	Male	63.0	88.0	78.0	77	2021	
4	Female	70.0	76.0	63.0	98	2019	
5	Male	68.0	78.0	79.0	79	2021	
6	Male	79.0	95.0	NaN	89	2018	
7	Male	77.0	NaN	70.0	84	2018	
8	Female	72.0	85.0	68.0	81	2021	
9	Female	62.0	76.0	74.0	98	2021	
10	Female	76.0	88.0	61.0	89	2021	
11	Female	72.0	81.0	78.0	82	2021	
12	Male	78.0	81.0	64.0	80	2018	
13	Female	62.0	76.0	NaN	94	2018	
14	Female	72.0	76.0	66.0	86	2021	
15	Female	79.0	91.0	67.0	79	2018	
16	Male	75.0	85.0	79.0	79	2020	
17	Female	74.0	91.0	73.0	88	2020	
18	Female	60.0	NaN	64.0	77	2021	
19	Male	68.0	78.0	77.0	90	2021	
20	Female	69.0	82.0	63.0	84	2018	
21	Male	72.0	86.0	NaN	85	2021	
22	Male	75.0	95.0	75.0	84	2018	
23	Male	74.0	86.0	66.0	93	2021	
24	Female	80.0	NaN	62.0	95	2020	
25	Female	68.0	83.0	79.0	79	2018	
26	Female	79.0	75.0	71.0	76	2021	
27	Female	73.0	79.0	70.0	100	2021	
28	Male	63.0	77.0	67.0	93	2020	
29	Female	78.0	84.0	74.0	91	2020	

In [49]:

```
df = pd.read_csv("Downloads/Students_Performance.csv")
df['Math_Score'] = df['Math_Score'].fillna(df['Math_Score'].median())
df
```

Out[49]:

	Gender	Math_Score	Reading_Score	Writing_Score	Placement_Score	Club_Join_Date	Placement_Offer_Coi
0	Male	69.0	95.0	71.0	93	2020	
1	Male	77.0	82.0	78.0	86	2019	
2	Male	76.0	90.0	70.0	83	2018	
3	Male	63.0	88.0	78.0	77	2021	
4	Female	70.0	76.0	63.0	98	2019	
5	Male	68.0	78.0	79.0	79	2021	
6	Male	79.0	95.0	NaN	89	2018	
7	Male	77.0	NaN	70.0	84	2018	
8	Female	74.0	85.0	68.0	81	2021	
9	Female	62.0	76.0	74.0	98	2021	
10	Female	76.0	88.0	61.0	89	2021	
11	Female	72.0	81.0	78.0	82	2021	
12	Male	78.0	81.0	64.0	80	2018	
13	Female	62.0	76.0	NaN	94	2018	
14	Female	74.0	76.0	66.0	86	2021	
15	Female	79.0	91.0	67.0	79	2018	
16	Male	75.0	85.0	79.0	79	2020	
17	Female	74.0	91.0	73.0	88	2020	
18	Female	60.0	NaN	64.0	77	2021	
19	Male	68.0	78.0	77.0	90	2021	
20	Female	69.0	82.0	63.0	84	2018	
21	Male	74.0	86.0	NaN	85	2021	
22	Male	75.0	95.0	75.0	84	2018	
23	Male	74.0	86.0	66.0	93	2021	
24	Female	80.0	NaN	62.0	95	2020	
25	Female	68.0	83.0	79.0	79	2018	
26	Female	79.0	75.0	71.0	76	2021	
27	Female	73.0	79.0	70.0	100	2021	
28	Male	63.0	77.0	67.0	93	2020	
29	Female	78.0	84.0	74.0	91	2020	

In [50]:

```
df = pd.read_csv("Downloads/Students_Performance.csv")
df['Math_Score'] = df['Math_Score'].fillna(df['Math_Score'].std())
df
```

Out[50]:	Gender	Math_Score	Reading_Score	Writing_Score	Placement_Score	Club_Join_Date	Placement_Offer_Coi
0	Male	69.000000	95.0	71.0	93	2020	
1	Male	77.000000	82.0	78.0	86	2019	
2	Male	76.000000	90.0	70.0	83	2018	
3	Male	63.000000	88.0	78.0	77	2021	
4	Female	70.000000	76.0	63.0	98	2019	
5	Male	68.000000	78.0	79.0	79	2021	
6	Male	79.000000	95.0	NaN	89	2018	
7	Male	77.000000	NaN	70.0	84	2018	
8	Female	6.101702	85.0	68.0	81	2021	
9	Female	62.000000	76.0	74.0	98	2021	
10	Female	76.000000	88.0	61.0	89	2021	
11	Female	72.000000	81.0	78.0	82	2021	
12	Male	78.000000	81.0	64.0	80	2018	
13	Female	62.000000	76.0	NaN	94	2018	
14	Female	6.101702	76.0	66.0	86	2021	
15	Female	79.000000	91.0	67.0	79	2018	
16	Male	75.000000	85.0	79.0	79	2020	
17	Female	74.000000	91.0	73.0	88	2020	
18	Female	60.000000	NaN	64.0	77	2021	
19	Male	68.000000	78.0	77.0	90	2021	
20	Female	69.000000	82.0	63.0	84	2018	
21	Male	6.101702	86.0	NaN	85	2021	
22	Male	75.000000	95.0	75.0	84	2018	
23	Male	74.000000	86.0	66.0	93	2021	
24	Female	80.000000	NaN	62.0	95	2020	
25	Female	68.000000	83.0	79.0	79	2018	
26	Female	79.000000	75.0	71.0	76	2021	
27	Female	73.000000	79.0	70.0	100	2021	
28	Male	63.000000	77.0	67.0	93	2020	
29	Female	78.000000	84.0	74.0	91	2020	

replacing missing values in forenoon column with minimum/maximum numberof that column

```
In [51]: df = pd.read_csv("Downloads/Students_Performance.csv")
df['Math_Score'] = df['Math_Score'].fillna(df['Math_Score'].min())
df
```

Out[51]:

	Gender	Math_Score	Reading_Score	Writing_Score	Placement_Score	Club_Join_Date	Placement_Offer_Coi
0	Male	69.0	95.0	71.0	93	2020	
1	Male	77.0	82.0	78.0	86	2019	
2	Male	76.0	90.0	70.0	83	2018	
3	Male	63.0	88.0	78.0	77	2021	
4	Female	70.0	76.0	63.0	98	2019	
5	Male	68.0	78.0	79.0	79	2021	
6	Male	79.0	95.0	NaN	89	2018	
7	Male	77.0	NaN	70.0	84	2018	
8	Female	60.0	85.0	68.0	81	2021	
9	Female	62.0	76.0	74.0	98	2021	
10	Female	76.0	88.0	61.0	89	2021	
11	Female	72.0	81.0	78.0	82	2021	
12	Male	78.0	81.0	64.0	80	2018	
13	Female	62.0	76.0	NaN	94	2018	
14	Female	60.0	76.0	66.0	86	2021	
15	Female	79.0	91.0	67.0	79	2018	
16	Male	75.0	85.0	79.0	79	2020	
17	Female	74.0	91.0	73.0	88	2020	
18	Female	60.0	NaN	64.0	77	2021	
19	Male	68.0	78.0	77.0	90	2021	
20	Female	69.0	82.0	63.0	84	2018	
21	Male	60.0	86.0	NaN	85	2021	
22	Male	75.0	95.0	75.0	84	2018	
23	Male	74.0	86.0	66.0	93	2021	
24	Female	80.0	NaN	62.0	95	2020	
25	Female	68.0	83.0	79.0	79	2018	
26	Female	79.0	75.0	71.0	76	2021	
27	Female	73.0	79.0	70.0	100	2021	
28	Male	63.0	77.0	67.0	93	2020	
29	Female	78.0	84.0	74.0	91	2020	

In [52]:

```
df = pd.read_csv("Downloads/Students_Performance.csv")
df['Math_Score'] = df['Math_Score'].fillna(df['Math_Score'].max())
df
```

Out [52]:	Gender	Math_Score	Reading_Score	Writing_Score	Placement_Score	Club_Join_Date	Placement_Offer_Coi
0	Male	69.0	95.0	71.0	93	2020	
1	Male	77.0	82.0	78.0	86	2019	
2	Male	76.0	90.0	70.0	83	2018	
3	Male	63.0	88.0	78.0	77	2021	
4	Female	70.0	76.0	63.0	98	2019	
5	Male	68.0	78.0	79.0	79	2021	
6	Male	79.0	95.0	NaN	89	2018	
7	Male	77.0	NaN	70.0	84	2018	
8	Female	80.0	85.0	68.0	81	2021	
9	Female	62.0	76.0	74.0	98	2021	
10	Female	76.0	88.0	61.0	89	2021	
11	Female	72.0	81.0	78.0	82	2021	
12	Male	78.0	81.0	64.0	80	2018	
13	Female	62.0	76.0	NaN	94	2018	
14	Female	80.0	76.0	66.0	86	2021	
15	Female	79.0	91.0	67.0	79	2018	
16	Male	75.0	85.0	79.0	79	2020	
17	Female	74.0	91.0	73.0	88	2020	
18	Female	60.0	NaN	64.0	77	2021	
19	Male	68.0	78.0	77.0	90	2021	
20	Female	69.0	82.0	63.0	84	2018	
21	Male	80.0	86.0	NaN	85	2021	
22	Male	75.0	95.0	75.0	84	2018	
23	Male	74.0	86.0	66.0	93	2021	
24	Female	80.0	NaN	62.0	95	2020	
25	Female	68.0	83.0	79.0	79	2018	
26	Female	79.0	75.0	71.0	76	2021	
27	Female	73.0	79.0	70.0	100	2021	
28	Male	63.0	77.0	67.0	93	2020	
29	Female	78.0	84.0	74.0	91	2020	

Filling null values in dataset

```
In [53]: df = pd.read_csv("Downloads/Students_Performance.csv")
mean_value=df['Math_Score'].mean()
df['Math_Score'].fillna(value=mean_value, inplace=True)
df
```


Out[53]:

	Gender	Math_Score	Reading_Score	Writing_Score	Placement_Score	Club_Join_Date	Placement_Offer_Coi
0	Male	69.0	95.0	71.0	93	2020	
1	Male	77.0	82.0	78.0	86	2019	
2	Male	76.0	90.0	70.0	83	2018	
3	Male	63.0	88.0	78.0	77	2021	
4	Female	70.0	76.0	63.0	98	2019	
5	Male	68.0	78.0	79.0	79	2021	
6	Male	79.0	95.0	NaN	89	2018	
7	Male	77.0	NaN	70.0	84	2018	
8	Female	72.0	85.0	68.0	81	2021	
9	Female	62.0	76.0	74.0	98	2021	
10	Female	76.0	88.0	61.0	89	2021	
11	Female	72.0	81.0	78.0	82	2021	
12	Male	78.0	81.0	64.0	80	2018	
13	Female	62.0	76.0	NaN	94	2018	
14	Female	72.0	76.0	66.0	86	2021	
15	Female	79.0	91.0	67.0	79	2018	
16	Male	75.0	85.0	79.0	79	2020	
17	Female	74.0	91.0	73.0	88	2020	
18	Female	60.0	NaN	64.0	77	2021	
19	Male	68.0	78.0	77.0	90	2021	
20	Female	69.0	82.0	63.0	84	2018	
21	Male	72.0	86.0	NaN	85	2021	
22	Male	75.0	95.0	75.0	84	2018	
23	Male	74.0	86.0	66.0	93	2021	
24	Female	80.0	NaN	62.0	95	2020	
25	Female	68.0	83.0	79.0	79	2018	
26	Female	79.0	75.0	71.0	76	2021	
27	Female	73.0	79.0	70.0	100	2021	
28	Male	63.0	77.0	67.0	93	2020	
29	Female	78.0	84.0	74.0	91	2020	

Filling a null values using replace() method

In [54]:

```
df = pd.read_csv("Downloads/Students_Performance.csv")
df.replace(to_replace = np.nan, value = -99)
```

Out [54]:	Gender	Math_Score	Reading_Score	Writing_Score	Placement_Score	Club_Join_Date	Placement_Offer_Coi
0	Male	69.0	95.0	71.0	93	2020	
1	Male	77.0	82.0	78.0	86	2019	
2	Male	76.0	90.0	70.0	83	2018	
3	Male	63.0	88.0	78.0	77	2021	
4	Female	70.0	76.0	63.0	98	2019	
5	Male	68.0	78.0	79.0	79	2021	
6	Male	79.0	95.0	-99.0	89	2018	
7	Male	77.0	-99.0	70.0	84	2018	
8	Female	-99.0	85.0	68.0	81	2021	
9	Female	62.0	76.0	74.0	98	2021	
10	Female	76.0	88.0	61.0	89	2021	
11	Female	72.0	81.0	78.0	82	2021	
12	Male	78.0	81.0	64.0	80	2018	
13	Female	62.0	76.0	-99.0	94	2018	
14	Female	-99.0	76.0	66.0	86	2021	
15	Female	79.0	91.0	67.0	79	2018	
16	Male	75.0	85.0	79.0	79	2020	
17	Female	74.0	91.0	73.0	88	2020	
18	Female	60.0	-99.0	64.0	77	2021	
19	Male	68.0	78.0	77.0	90	2021	
20	Female	69.0	82.0	63.0	84	2018	
21	Male	-99.0	86.0	-99.0	85	2021	
22	Male	75.0	95.0	75.0	84	2018	
23	Male	74.0	86.0	66.0	93	2021	
24	Female	80.0	-99.0	62.0	95	2020	
25	Female	68.0	83.0	79.0	79	2018	
26	Female	79.0	75.0	71.0	76	2021	
27	Female	73.0	79.0	70.0	100	2021	
28	Male	63.0	77.0	67.0	93	2020	
29	Female	78.0	84.0	74.0	91	2020	

Deleting null values using dropna() method

```
In [55]: # To drop rows with at least 1 null value
df = pd.read_csv("Downloads/Students_Performance.csv")
df.dropna()
```

Out[55]:

	Gender	Math_Score	Reading_Score	Writing_Score	Placement_Score	Club_Join_Date	Placement_Offer_Coi
0	Male	69.0	95.0	71.0	93	2020	
1	Male	77.0	82.0	78.0	86	2019	
2	Male	76.0	90.0	70.0	83	2018	
3	Male	63.0	88.0	78.0	77	2021	
4	Female	70.0	76.0	63.0	98	2019	
5	Male	68.0	78.0	79.0	79	2021	
9	Female	62.0	76.0	74.0	98	2021	
10	Female	76.0	88.0	61.0	89	2021	
11	Female	72.0	81.0	78.0	82	2021	
12	Male	78.0	81.0	64.0	80	2018	
15	Female	79.0	91.0	67.0	79	2018	
16	Male	75.0	85.0	79.0	79	2020	
17	Female	74.0	91.0	73.0	88	2020	
19	Male	68.0	78.0	77.0	90	2021	
20	Female	69.0	82.0	63.0	84	2018	
22	Male	75.0	95.0	75.0	84	2018	
23	Male	74.0	86.0	66.0	93	2021	
25	Female	68.0	83.0	79.0	79	2018	
26	Female	79.0	75.0	71.0	76	2021	
27	Female	73.0	79.0	70.0	100	2021	
28	Male	63.0	77.0	67.0	93	2020	
29	Female	78.0	84.0	74.0	91	2020	

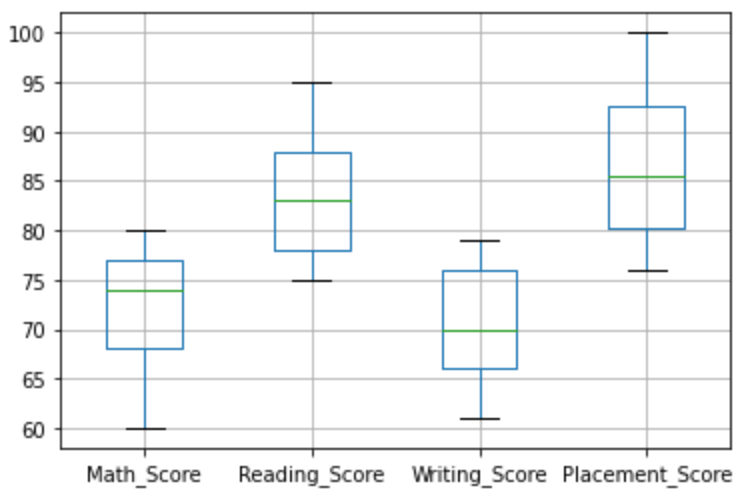
Identification and Handling of Outliers

```
In [60]: # Load dataset
df = pd.read_csv("Downloads/Students_Performance.csv")

# Remove any extra spaces or tabs from column names
df.columns = df.columns.str.strip()

# Define the corrected column list
col = ['Math_Score', 'Reading_Score', 'Writing_Score', 'Placement_Score']

# Draw the boxplot
df.boxplot(column=col)
plt.show()
```



```
In [67]: # We can now print the outliers for each column with reference to the box
plot.
print(np.where(df['Math_Score']>80))
print(np.where(df['Reading_Score']<75))
print(np.where(df['Writing_Score']>80))
```

```
(array([], dtype=int64),)
(array([], dtype=int64),)
(array([], dtype=int64),)
```

```
In [73]: df = pd.read_csv("Downloads/Students_Performance.csv")

# Remove any extra spaces/tabs from column names
df.columns = df.columns.str.strip()

# Calculate mean and standard deviation for Math_Score
mean_math = df["Math_Score"].mean()
std_math = df["Math_Score"].std()

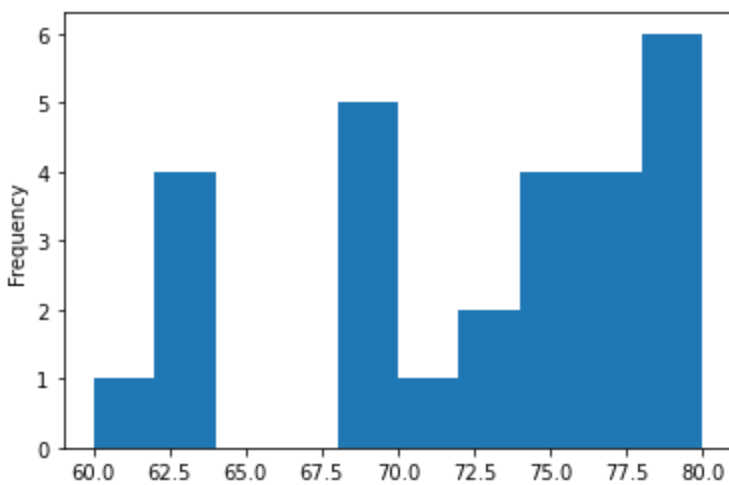
# Compute Z-scores for Math_Score
df["Zscore"] = (df["Math_Score"] - mean_math) / std_math

# Display the first few rows
print(df[["Math_Score", "Zscore"]].head())
```

	Math_Score	Zscore
0	69.0	-0.491666
1	77.0	0.819443
2	76.0	0.655555
3	63.0	-1.474998
4	70.0	-0.327777

```
In [69]: # Observe the histogram for math_score variable.
import matplotlib.pyplot as plt
df['Math_Score'].plot(kind = 'hist')
```

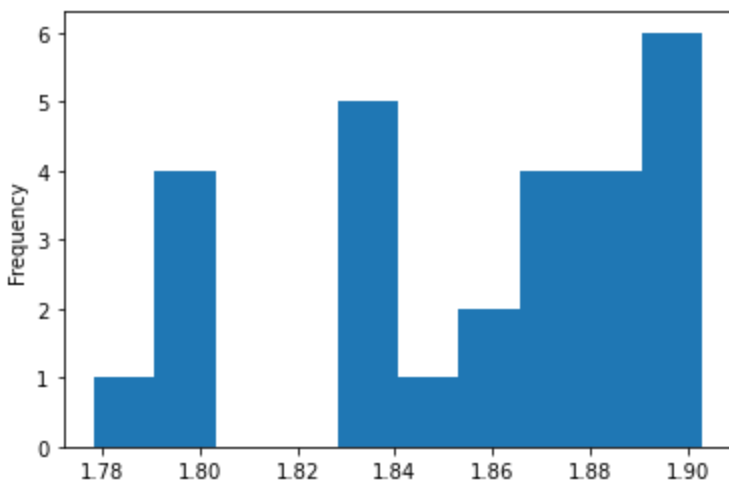
Out[69]: <AxesSubplot:ylabel='Frequency'>



```
In [70]: # Convert the variables to logarithm at the scale 10.  
df['log_math'] = np.log10(df['Math_Score'])
```

```
In [71]: # Observe the histogram for math_score variable.  
df['log_math'].plot(kind = 'hist')
```

Out[71]: <AxesSubplot:ylabel='Frequency'>



In []: